

RV College of Engineering*, Bengaluru – 59
Department of Computer Science and Engineering
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING (IS353IA)

Synopsis

TITLE: Precision Agriculture using Machine Learning

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Introduction: The "Traffic Monitoring System" builds upon object detection and tracking technologies to address real-world challenges in traffic management. By integrating YOLOv8 for object detection and Deep SORT for efficient tracking, this system provides a robust solution to monitor, analyze, and manage road traffic. With the increasing number of vehicles on the road, traffic management has become a critical challenge. Utilizing advanced Machine Learning techniques to detect, track, and count vehicles, estimate their speeds, and identify speed limit violations, ensuring proactive traffic management and safety. It detects vehicles, tracks their movements, estimates speeds, and identifies speed limit violations, enabling safer and more efficient transportation systems.

Existing System: Existing traffic monitoring systems rely on basic detection methods or manual interventions or basic camera systems, which are often inefficient and lack the precision to handle real-time traffic complexities.

Proposed System: This project advances object detection and tracking techniques by applying them to traffic monitoring scenarios. Features include: Detect and track vehicles in real-time.

1. **Vehicle Detection and Classification:** Leveraging YOLOv8's state-of-the-art object detection capabilities to identify vehicles such as cars, trucks, buses, and motorcycles.
2. **Speed Estimation:** Using bounding box movements and time-based calculations for speed measurement.
3. **Speed Limit Violation Detection:** Real-time alerts for vehicles exceeding a defined speed threshold.
4. **Counting and Categorization:** Counting vehicles entering or leaving specific regions of interest.

Data Sources:

1. **Pre-trained YOLOv8 Models:** Trained on the COCO dataset, featuring 80 object categories.
2. **Custom Traffic Video Datasets:** Real-world video footage for validation and testing of the system.

Technical Implementation:

1. **Machine Learning Models:** YOLOv8 models for object detection, integrated with a centroid tracking algorithm for vehicle tracking.
2. **Frontend:** A dashboard for displaying real-time results and analytics.
3. **Speed Calculation:** Speed is estimated by measuring the time vehicles take to traverse predefined regions.
4. **Backend:** Python-based implementation using frameworks like Flask to process and serve predictions.
5. **Deployment:** Hosted on cloud platforms (e.g., AWS, Heroku) for scalability and real-time monitoring.

Societal Concern: This project aims to enhance road safety by providing tools for efficient traffic monitoring and management. By detecting speed violations and providing real-time traffic insights, it contributes to reduced accidents and improved commuter safety.

Conclusion: By extending object detection and tracking technologies, the "Traffic Monitoring System" transforms them into practical applications for real-world challenges and its leverages cutting-edge AI models to address critical traffic management issues. Its implementation promotes safety, efficiency, and smarter urban traffic management.